# MTH636 – Game Theory

## Lecture 13: The Indifference Principle and Finding NE

## 1 The Indifference Principle

**Theorem: The Indifference Principle**

Suppose $\sigma^*$ is a Nash Equilibrium. For any player $i$, if there are two pure strategies $s_i^1, s_i^2 \in S_i$ that are played with strictly positive probability (i.e., $\sigma_i^*(s_i^1) > 0$ and $\sigma_i^*(s_i^2) > 0$), then:

$$U_i(s_i^1, \sigma_{-i}^*) = U_i(s_i^2, \sigma_{-i}^*)$$

**Proof Idea:** Suppose for contradiction that $U_i(s_i^1, \sigma_{-i}^*) > U_i(s_i^2, \sigma_{-i}^*)$. Player $i$ could define a new mixed strategy $\bar{\sigma}_i$ by taking all the probability mass assigned to $s_i^2$ and shifting it to $s_i^1$:

$$\bar{\sigma}_i(s_i) = \begin{cases} \sigma_i^*(s_i) & \text{if } s_i \neq s_i^1, s_i^2 \\ 0 & \text{if } s_i = s_i^2 \\ \sigma_i^*(s_i^1) + \sigma_i^*(s_i^2) & \text{if } s_i = s_i^1 \end{cases}$$

By evaluating the expected utility $U_i(\bar{\sigma}_i, \sigma_{-i}^*)$, one can show that Player $i$ obtains a strictly higher expected payoff than $U_i(\sigma_i^*, \sigma_{-i}^*)$, which contradicts the assumption that $\sigma^*$ is a Nash Equilibrium.

**Note**

**Corollary:** For any Nash Equilibrium $\sigma^*$, the expected utility $U_i(\sigma^*)$ is exactly equal to the expected utility of **any** pure strategy $s_i$ in its support:

$$U_i(\sigma^*) = U_i(s_i, \sigma_{-i}^*) \quad \forall s_i \in S_i \text{ such that } \sigma_i^*(s_i) > 0$$

## 2 Computing Mixed NE using Indifference

**Example: Computing a Mixed NE**

Consider the following game:

| $P_1 \setminus P_2$ | L ($y$) | R ($1-y$) |
|---|---|---|
| T ($x$) | (1, −1) | (0, 2) |
| B ($1-x$) | (0, 1) | (2, 0) |

There is no pure strategy NE, nor is there an NE where one player plays a pure strategy and the other plays a proper mixed strategy. Thus, any NE must have $0 < x < 1$ and $0 < y < 1$.

Using the Indifference Principle for Player 1:

$$\begin{aligned} U_1(T, y) &= U_1(B, y) \\ 1(y) + 0(1-y) &= 0(y) + 2(1-y) \\ y = 2 - 2y \implies 3y = 2 \implies y &= \frac{2}{3} \end{aligned}$$

Using the Indifference Principle for Player 2:

$$\begin{aligned} U_2(x, L) &= U_2(x, R) \\ -1(x) + 1(1-x) &= 2(x) + 0(1-x) \\ 1 - 2x = 2x \implies 4x = 1 \implies x &= \frac{1}{4} \end{aligned}$$

The unique Nash Equilibrium is $\sigma_1^* = (1/4, 3/4)$ and $\sigma_2^* = (2/3, 1/3)$.

## 3 Algorithm for Finding All NE of a Two-Player Game

To find all Nash Equilibria of a general two-player finite game, we can systematically search over all possible **supports**. Let $S_1$ and $S_2$ be the finite sets of pure strategies, with $|S_1| = m_1$ and $|S_2| = m_2$.

For every possible subset $Y_1 \subseteq S_1$ and $Y_2 \subseteq S_2$ (there are $(2^{m_1} - 1) \times (2^{m_2} - 1)$ such pairs of supports), we check if there exists an NE $(\sigma_1^*, \sigma_2^*)$ such that $\text{supp}(\sigma_1^*) = Y_1$ and $\text{supp}(\sigma_2^*) = Y_2$.

Such a profile is an NE if and only if it satisfies the following system of linear equations and inequalities:

1. **Indifference within the support (Player 1):**
   For some fixed $s_1^o \in Y_1$,
   $$\sum_{s_2 \in Y_2} \sigma_2^*(s_2) u_1(s_1^o, s_2) = \sum_{s_2 \in Y_2} \sigma_2^*(s_2) u_1(s_1, s_2) \quad \forall s_1 \in Y_1$$

2. **Indifference within the support (Player 2):**
   For some fixed $s_2^o \in Y_2$,
   $$\sum_{s_1 \in Y_1} \sigma_1^*(s_1) u_2(s_1, s_2^o) = \sum_{s_1 \in Y_1} \sigma_1^*(s_1) u_2(s_1, s_2) \quad \forall s_2 \in Y_2$$

3. **No profitable deviation outside the support (Player 1):**
   $$\sum_{s_2 \in Y_2} \sigma_2^*(s_2) u_1(s_1^o, s_2) \geq \sum_{s_2 \in Y_2} \sigma_2^*(s_2) u_1(s_1, s_2) \quad \forall s_1 \notin Y_1$$

4. **No profitable deviation outside the support (Player 2):**
   $$\sum_{s_1 \in Y_1} \sigma_1^*(s_1) u_2(s_1, s_2^o) \geq \sum_{s_1 \in Y_1} \sigma_1^*(s_1) u_2(s_1, s_2) \quad \forall s_2 \notin Y_2$$

5. **Valid probabilities (sum to 1):**

$$\sum_{s_1 \in Y_1} \sigma_1^*(s_1) = 1, \quad \sum_{s_2 \in Y_2} \sigma_2^*(s_2) = 1$$

6. **Valid probabilities (strictly positive on support):**

$$\sigma_1^*(s_1) > 0 \quad \forall s_1 \in Y_1, \quad \sigma_2^*(s_2) > 0 \quad \forall s_2 \in Y_2$$

7. **Zero probability outside support:**

$$\sigma_1^*(s_1) = 0 \quad \forall s_1 \notin Y_1, \quad \sigma_2^*(s_2) = 0 \quad \forall s_2 \notin Y_2$$

## 3.1 Simplifying the Search via Dominance

Before applying the exhaustive search, one should iteratively eliminate strictly dominated strategies to reduce the sizes of $S_1$ and $S_2$.

**Example: Strict Dominance by a Mixed Strategy**

| $P_1 \setminus P_2$ | L | C | R |
|---|---|---|---|
| T | (6, 2) | (0, 6) | (4, 4) |
| M | (2, 12) | (4, 3) | (2, 5) |
| B | (0, 6) | (10, 0) | (2, 2) |

Notice that the pure strategy $M$ for Player 1 is strictly dominated by the mixed strategy $\sigma_1 = \left(\frac{1}{2}T, 0M, \frac{1}{2}B\right)$:

- Against L: $\frac{1}{2}(6) + \frac{1}{2}(0) = 3 > 2$
- Against C: $\frac{1}{2}(0) + \frac{1}{2}(10) = 5 > 4$
- Against R: $\frac{1}{2}(4) + \frac{1}{2}(2) = 3 > 2$

Since $M$ is strictly dominated, it can never be in the support of any Nash Equilibrium, and we can immediately eliminate it to simplify the game to a $2 \times 3$ game.